

Cayman Roden

AI Engineer | Production RAG · Agents · Evals · FastAPI · Python

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PROFESSIONAL SUMMARY

AI Engineer building production Python-based RAG, agent workflows, evals, and backend systems from prototype to deployment. Shipped a Claude-powered lead qualification platform that processed 500+ leads, built DocExtract with 94.6% eval accuracy, and published mcp-server-toolkit to PyPI. Strongest edge: turning AI prototypes into tested, deployable systems with direct stakeholder ownership. Open source contributor to LiteLLM, FastAPI, and pgvector-python. 9,900+ automated tests across public production repos.

Live: docextract-demo.streamlit.app | PyPI: mcp-server-toolkit

TECHNICAL SKILLS

LLM and GenAI: Claude API (tool use, streaming, multi-turn) | OpenAI API | Gemini API | LangGraph | Prompt Engineering | Model Evaluation | RAGAS | LLM-as-Judge **RAG and Retrieval:** Retrieval-Augmented Generation (RAG) | pgvector | Embedding Models | Semantic Search | BM25 | Reciprocal Rank Fusion | Agentic RAG | ReAct Reasoning | Semantic Caching | Document Chunking **Agentic AI and MCP:** Agentic AI | Multi-Agent Orchestration | Model Context Protocol (MCP) | MCP Server | A2A Adapter | Tool Use / Function Calling | Structured Output | Agent State Management | Circuit Breakers | Per-Agent Model Routing | Guardrails **Reliability and Delivery:** Prompt Versioning | Evaluation Pipelines | Golden Dataset Curation | Cost Governance | GitHub Actions CI/CD | OpenTelemetry | Observability **Backend:** Python (asyncio, FastAPI, pytest, Pydantic, SQLAlchemy, Alembic) | TypeScript | SQL | PostgreSQL | Redis | Docker | REST API **Testing:** TDD | Unit Testing | Integration Testing | pytest-asyncio | Load Testing (Locust) | Adversarial Testing | Hallucination Detection **Methods:** Agile | Code Review | Technical Documentation | API Design

OPEN SOURCE CONTRIBUTIONS

- **LiteLLM** (BerriAI/litellm, 27K+ stars): Open PR #24551 - typed exception mapping for Router fallback, surfaces AuthenticationError, RateLimitError, NotFoundError distinctly instead of swallowing as generic Exception
- **FastAPI** (80K+ stars): Open PR #15217 - BackgroundTasks interaction warning for dependency injection chains
- **pgvector-python**: Open PR #151 - async SQLAlchemy + HNSW index documentation covering connection pooling and performance-critical index selection

EXPERIENCE

AI Engineer (Contract) - Remote - June 2022 - Present

- Built and ran 3 Claude-powered SMS bots for real estate lead qualification, processing 500+ leads with sub-500ms responses, bilingual EN/ES support, and zero downtime across a 3-month production run. Implemented FastAPI webhooks, Redis-backed state/rate limiting, eval harnesses, and structured decision logging. 1,700+ tests.
- Built DocExtract, a production document-intelligence pipeline with 94.6% extraction accuracy on a 28-fixture eval set including 12 adversarial cases and 4 prompt-injection attacks. Implemented hybrid retrieval, ReAct-style tool use, semantic caching, circuit-breaker model fallback, and CI eval gates. 1,185 tests.
- Architected a multi-agent orchestration platform with a 3-tier cache that cut token usage 89% per workflow (93K to 7.8K). Added golden evals, prompt rollback, adversarial testing, and per-agent cost governance across a 6,700-test codebase.
- Published mcp-server-toolkit to PyPI with 9 pre-built MCP servers, caching, rate limiting, auth middleware, and test utilities, reducing LLM tool integration from days to a single import. 412 tests, 88% coverage.
- Built a real-time voice AI pipeline with Silero-ONNX VAD, Deepgram STT/TTS, Claude reasoning, and barge-in support, achieving sub-300ms time-to-first-byte in streaming interactions. 171 tests.

Previous Career | Operations Management - 2012 - 2022

10 years client-facing operations; stakeholder communication and scope management applied to technical delivery.

PROJECTS

Jorge Real Estate AI - Production LLM Application

FastAPI - Redis - GoHighLevel - Claude API

Three Claude-powered SMS bots handling lead qualification for a real estate business. Multi-turn conversation state, webhook verification, Redis-backed rate limiting, bilingual EN/ES support, and human handoff flows. 500+ leads processed with zero downtime across a 3-month production run. 1,700+ tests.

DocExtract AI - Production RAG Pipeline

FastAPI - ARQ - pgvector - Claude API - Streamlit

Async document processing with hybrid retrieval (BM25 + cosine semantic search + RRF), citation-aware answers, agentic ReAct reasoning, semantic caching, and evaluation pipelines with hallucination detection. Includes reference deployment configs, eval scripts, and a live demo. 1,185 tests, 87%+ coverage. Live: docextract-demo.streamlit.app

EnterpriseHub - Multi-Agent Orchestration

FastAPI - PostgreSQL - Redis - LangGraph - Claude API - Prometheus - Grafana

22-agent mesh with weighted routing, \$100/hr emergency shutdown, and 89% token cost reduction via a 3-tier cache. Strongest signals are eval-driven development, adversarial testing, prompt version rollback, and per-agent cost governance. 6,700 tests.

mcp-server-toolkit - Published PyPI Package

Python - MCP SDK - OpenTelemetry

9 pre-built MCP servers with A2A adapter, auto-caching, rate limiting, auth middleware. MCPTestClient for testing without live API keys. 412 tests, 88% coverage. pip install mcp-server-toolkit

CERTIFICATIONS

- IBM Generative AI Engineering with PyTorch, LangChain & Hugging Face Professional Certificate — 144h
- DeepLearning.AI Deep Learning Specialization — 120h
- Microsoft AI & ML Engineering Professional Certificate — 75h
- Duke University Large Language Model Operations (LLMOps) Specialization — 48h
- IBM RAG and Agentic AI: Build Next-Gen AI Systems Professional Certificate — 24h
- Google Cloud Generative AI Leader Certificate — 25h
- Claude Code in Action — Anthropic — 3h

21 total certifications · 1,831 hours · Google, IBM, Microsoft, DeepLearning.AI, Duke, Vanderbilt, Anthropic